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Project repository: <https://github.com/amaliza-shal/Time-Series-Forecasting1.git>

VideoLink: <https://www.awesomescreenshot.com/video/56694127?key=838092b6bca35d22b93c82fedcacb78f>

Mobile Network Traffic Forecasting on the Milan CDR Grid

CASE STUDY: One-step-ahead forecasting of Internet traffic for December 16–22, 2013 using SARIMA, LSTM and TCN

Research question. How do different sequential models compare for one-step-ahead mobile network traffic forecasting, and how does their performance vary across geographical areas with different traffic characteristics?

Introduction.

Mobile traffic varies considerably with time and from one location to another. Short-term forecasts are useful for capacity planning and for making operational decisions since they show when demand is likely to increase or decrease. In this study the forecasting problem is examined by using Internet activity from the Telecom Italia Milan CDR network. Instead of considering the whole city as a single series, the experiment looks at the three most busy geographical squares and investigates whether different families of sequential models react differently when the underlying traffic patterns change.

The study compares a classical statistical model (SARIMA-style dynamic harmonic regression with ARIMA errors), a recurrent neural network (LSTM), and a temporal convolutional network (TCN). The comparison uses the same target, the same chronological split, and one-step-ahead walk-forward evaluation. The objective is not only to report error values, but to connect those errors to the observed traffic structure, the way each model represents time, computational cost, and the limitations of the experimental design

Study scope. The dataset covers Milan from November 1, 2013 to January 1, 2014 at 10-minute resolution across 10,000 grid squares. The held-out test period is December 16–22, 2013, containing 1,008 ten-minute steps. The report therefore evaluates a specific historical

forecasting setting rather than claiming universal performance across all mobile networks.

Key findings in brief. The selected SARIMA configuration has the lowest MAE, MAPE and RMSE on all three busiest squares in this experiment. The neural models are nevertheless relatively close on two squares, and the analysis shows that traffic characteristics including strong daily/weekly repetition and different weekend behaviour are important to interpreting the results.

Related Work and Dataset

Related work. It Describes the multi-source Milan dataset and provides the provenance for the Telecom Italia activity data used here and compares LSTM and ARIMA for cellular traffic prediction and report that LSTM can benefit from large, fine trained training data while ARIMA can remain competitive at lower cost. This motivated a direct comparison between a classical time-series model and a recurrent neural network rather than selecting only one family.

Dataset. The Milan Telecom Italia CDR data divide the city into a 100×100 grid of 10,000 squares. For each square and 10-minute interval, the raw records contain SMS-in, SMS-out, call-in, call-out and Internet activity by country code. The forecasting target is the internet field, aggregated over all country-code rows sharing the same square and timestamp.

Raw schema. The files are tab-separated and have no header. The fields are square_id, timestamp_ms, country_code, sms_in, sms_out, call_in, call_out, internet. Many cells are empty because each row contains only the activity types observed for that country code and time. There is one raw file per day; the report's data audit confirmed that the December 16–22 test-week files were present.

Data scale. The raw dataset is approximately 20 GB across 62 daily files; one day is about 340 MB and 5.3 million rows. Where only being able to store it on my laptop i had to use flash disks rather than easily downloading the data which became an overcomable challenge , and also a full in-memory load inappropriate for my laptop with about 7.7 GB RAM.

Data Preparation and Memory Management

Two-pass loading strategy. The raw files were processed sequentially rather than concatenated in memory. In Pass 1, only square_id and internet were read in chunks of 2,000,000 rows, using int32/float32 where appropriate. Running totals identified the three busiest squares. In Pass 2, only square_id, timestamp_ms and internet were read, and the three selected squares plus required comparison squares 4159 and 4556 were extracted into compact time series.

This approach intentionally trades some extra disk I/O for much lower memory pressure. A single-pass strategy would have needed to retain many candidate areas before their final ranking was known. Garbage collection was used between stages, and the final extracted series were stored in a small Parquet representation suitable for modelling.

Stage	RSS MB	Time s	Δ baseline
Baseline	204.0	—	+0.0
Naive full file	500.4	3.2	+296.4
Chunked Pass 1	217.6	136.4	+13.6
Chunked Pass 2	232.2	154.8	+28.2

Memory evidence. The baseline RSS was 204.0 MB. Reading one complete day naively raised RSS to 500.4 MB, an increase of 296.4 MB. The chunked ranking pass ended at 217.6 MB and the extraction pass at 232.2 MB. These are post-stage RSS measurements rather than peak-memory measurements, so the true peak may have been higher.

For one typical day, reading all eight columns with default datatypes produced a DataFrame of about 296 MB. Reading only the two ranking columns with compact datatypes reduced that to about 37 MB (87% lower), while the three-column extraction pass used about 74 MB (75% lower). The measurements therefore provide direct evidence that column selection and datatype optimisation, combined with chunking, made the data manageable on the available hardware.

Data-quality checks. The inspection process verified the file structure, timestamps, date range, Square IDs, sampling interval and required test-week coverage before modelling. The workflow also checked missing and duplicate records and computed the traffic ranking used in the experiment. No modelling stage was started until these structural checks had been completed.

Reproducibility. The preparation is scriptable raw data that remained outside the modelling notebook, while the extracted series and ranking tables were small enough to support repeatable experiments. The raw dataset itself should not be committed to GitHub because of its size.

Exploratory Analysis

Traffic distribution. Total Internet traffic per square is strongly right-skewed. Most squares carry relatively little traffic, while a small number carry much more. The busiest square, 5161, has approximately 1.27×10^7 total Internet traffic units. A logarithmic view makes the long tail easier to inspect.

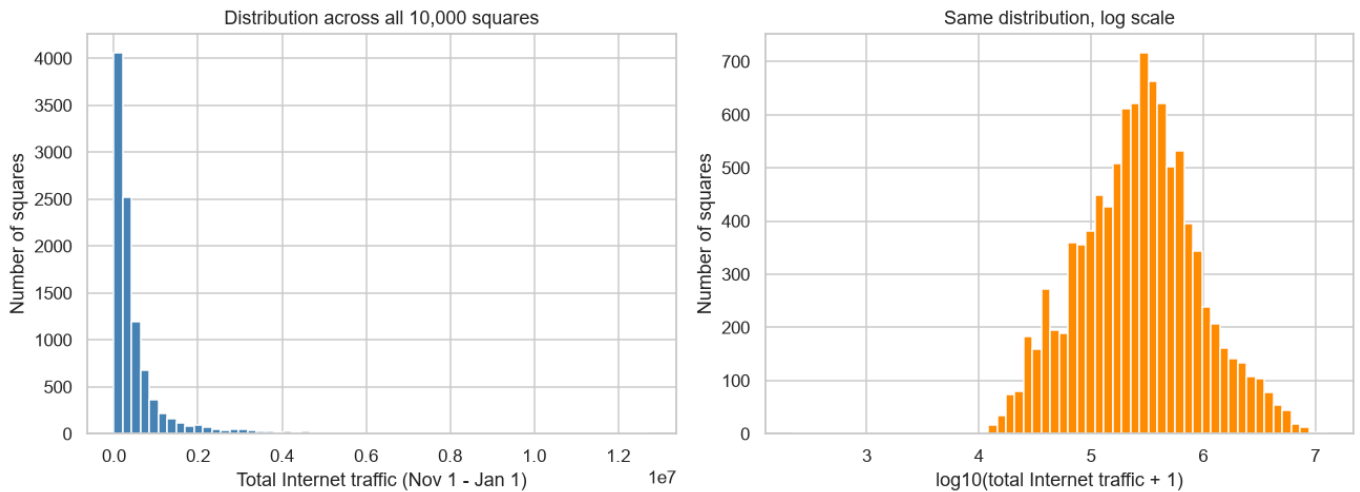


Figure 1. Distribution of total Internet traffic across the 10,000 geographical squares, shown on the original and logarithmic scales.

Ra nk	Square	Total Internet
1	5161	12,740,060
2	5059	11,170,854
3	5259	10,485,780
4	5061	9,584,334
5	5258	8,707,440
6	5159	8,703,873
7	6064	8,675,343
8	4855	8,491,044
9	4856	8,230,212
10	5262	8,163,357

The three highest-traffic squares are **5161, 5059 and 5259**. They are the focus of the final forecasting comparison. Squares 4159 and 4556 are retained because the assignment requires their inclusion in the first-two-weeks exploratory analysis.

Spatial concentration. The 100 busiest squares (1% of the grid) account for about 11% of total traffic, while the 1,000 busiest squares (10%) account for about 48%. The top three are spatially close: 5059 is at approximately (50,58), 5161 at (51,60), and 5259 at (52,58), using the row-wise grid convention.

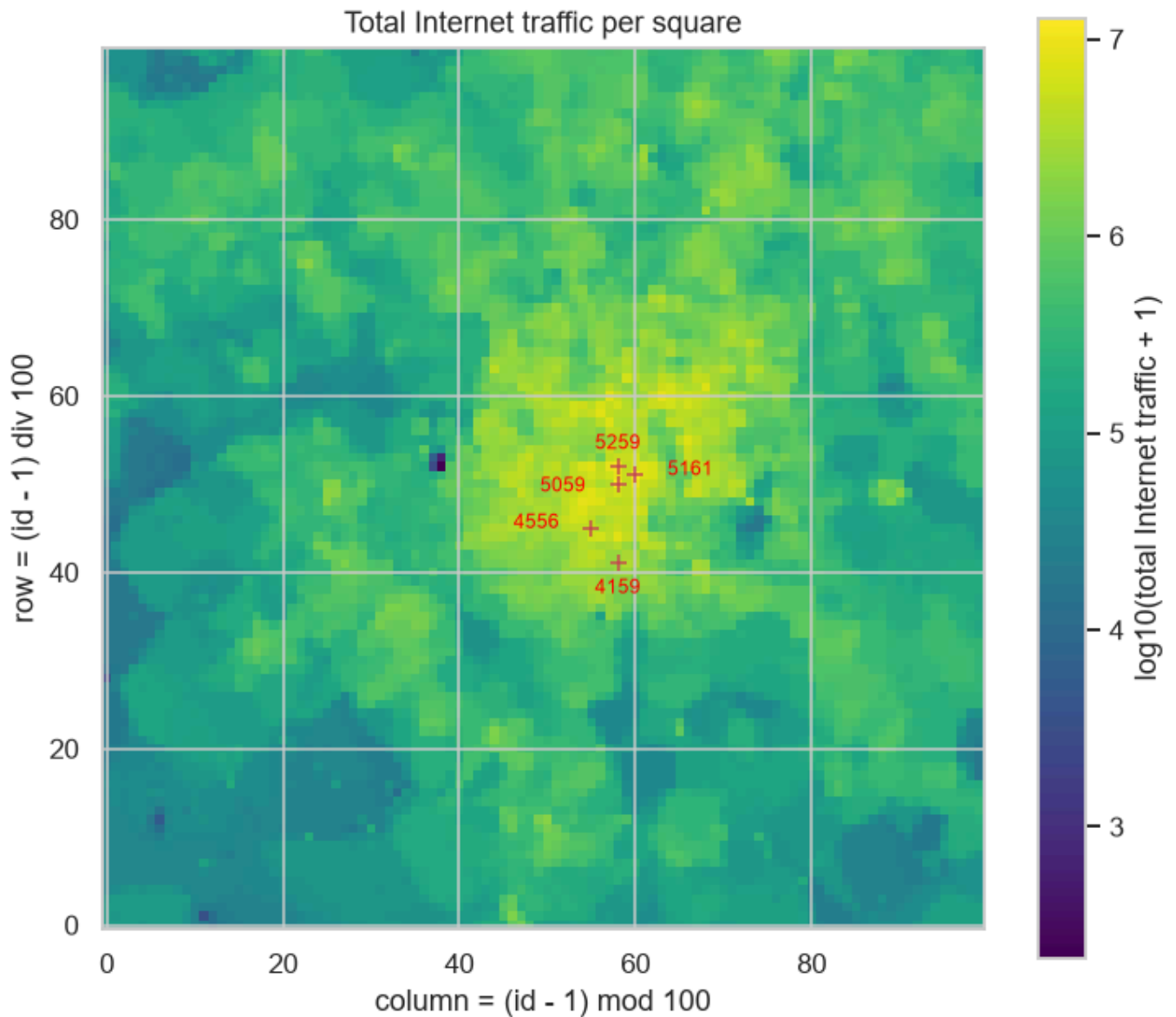


Figure 2. Spatial distribution of total Internet traffic and marked areas from the original analysis.

Spatial proximity does not guarantee the same temporal pattern. This becomes important later: 5161 and 5259 are close on the grid but have a full-period correlation of only 0.47, while 5259 and 4159 reach 0.90 despite their different traffic scales.

Exploratory Data Analysis: Temporal Behaviour

First two weeks. November 1–14 shows a clear daily cycle in all five required areas, with overnight troughs and daytime peaks. Weekdays (Nov 4–8 and Nov 11–14) are more similar across the three busiest squares, while weekends reveal much larger differences. Square 5161 reached an unusually high peak of about 8,044 on Saturday Nov 2. Square 5259, by contrast, is much lower on weekends.

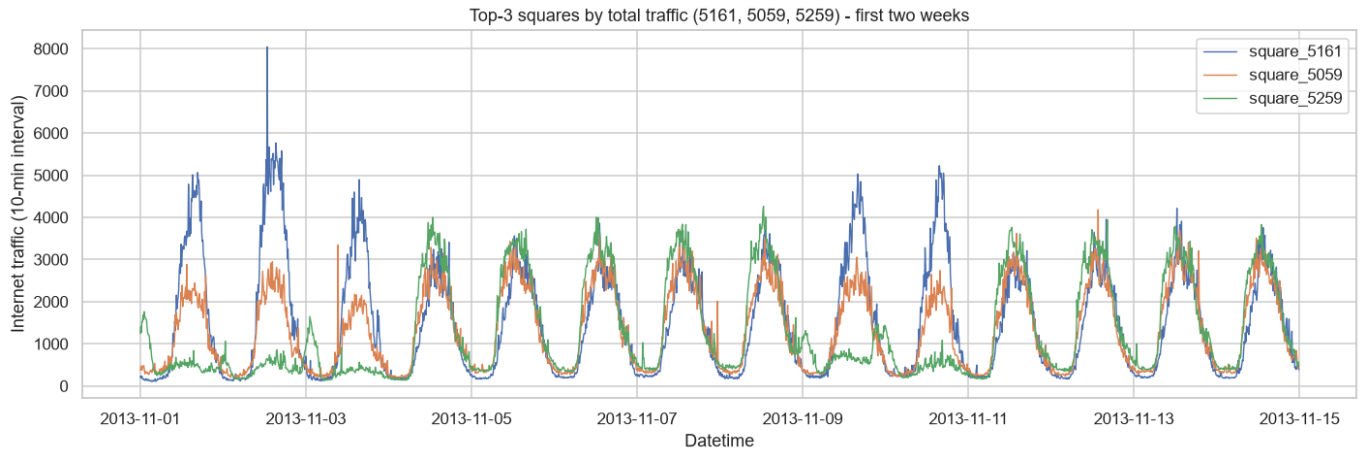


Figure 3. First-two-weeks temporal comparison for the top three squares and the required lower-volume squares.

Square	Mean	Max	CV		$\kappa/\text{weekday}$	Weekend	Max time
5161	1484	8044	0.88	5.4	1381	1740	Nov 02 12:50
5059	1331	4183	0.72	3.1	1424	1098	Nov 12 13:40
5259	1292	4263	0.88	3.3	1599	527	Nov 08 12:40
4159	311	851	0.61	2.7	361	187	Nov 04 17:10
4556	592	1869	0.43	3.2	577	627	Nov 09 21:10

Cross-area correlation. Correlations for the full 10-minute series are: 5161–5059 = 0.88; 5161–5259 = 0.47; 5161–4159 = 0.49; 5161–4556 = 0.39; 5059–5259 = 0.76; 5059–4159 = 0.77; 5059–4556 = 0.40; 5259–4159 = 0.90; 5259–4556 = 0.30; 4159–4556 = 0.39.

Before December 16, the 5161–5259 relationship changes strongly by day type: about 0.84 on weekdays but approximately -0.05 on weekends. The 5161–5059 relationship remains high at about 0.93 on weekdays and 0.95 on weekends. These observations support evaluating several areas rather than treating one square as representative of the whole grid.

Additional Temporal Analysis and Evidence for Model Choice.

STL decomposition. For the highest-traffic square, 5161, STL with a daily period of 144 observations gives seasonal strength 0.858 and trend strength 0.238. The daily seasonal component is therefore large and regular. The trend stays roughly between 1,300 and 1,750 until around December 22 and then drops sharply near the end of the year. The residual contains large negative deviations around December 25–26, when the daily peaks are much lower than the usual pattern.

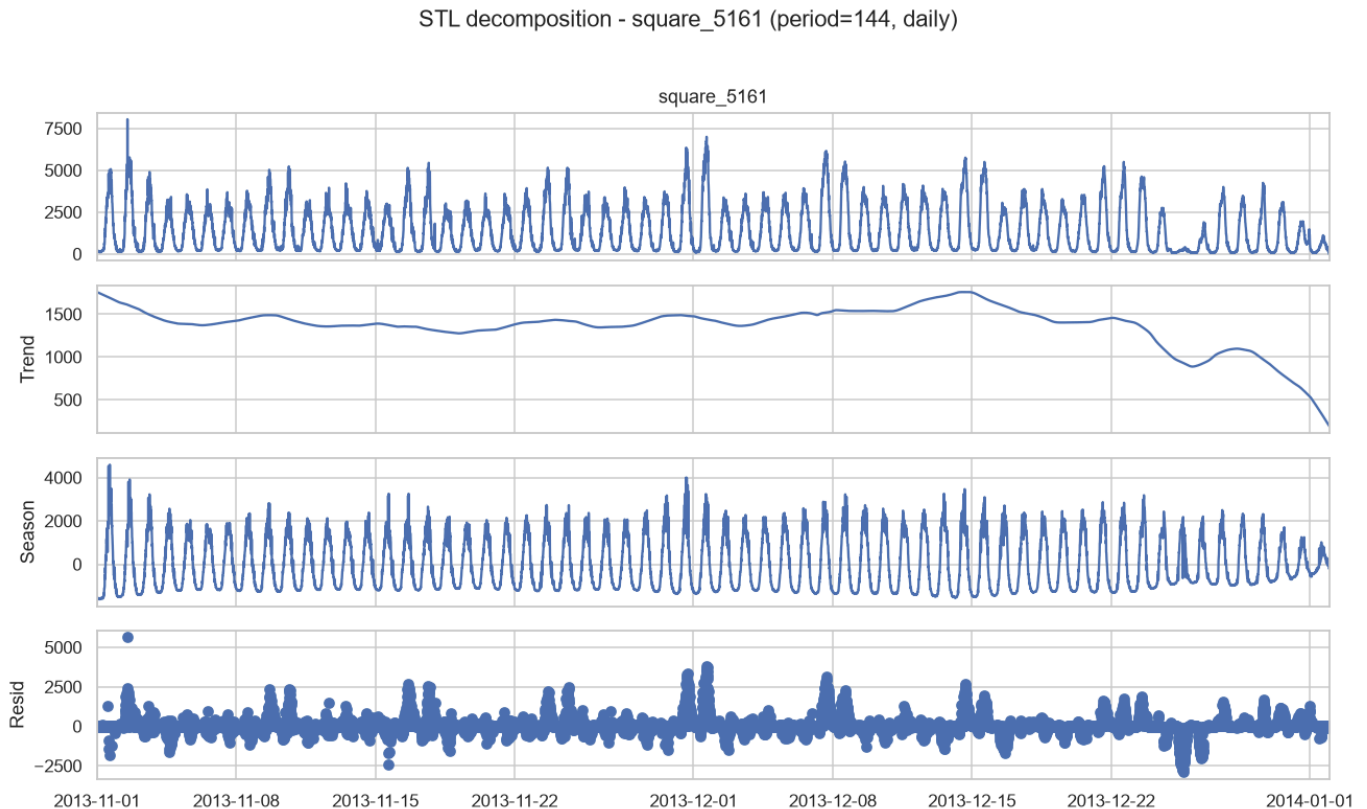


Figure 4. STL decomposition of Square 5161 over the full period; the daily seasonal component dominates much of the variation.

Autocorrelation and stationarity. The ACF for 5161 remains high at daily and weekly lags: approximately 0.878 at lag 144, 0.770 at 288, 0.741 at 432, and 0.838 at lag 1,008 (one week). The PACF is very large at lag 1 (0.987), falls to about 0.259 at lag 2, and is much smaller afterward. These values show strong repeated temporal structure.

The ADF statistic is -19.028 with $p < 0.001$, using 36 lags and 8,891 observations. The unit-root null is rejected under the test, although the series still contains strong daily and weekly dependence. In the modelling stage, the selected ARIMA order nevertheless included $d = 1$; this was an empirical model-selection result rather than a direct claim that the raw series was non-stationary.

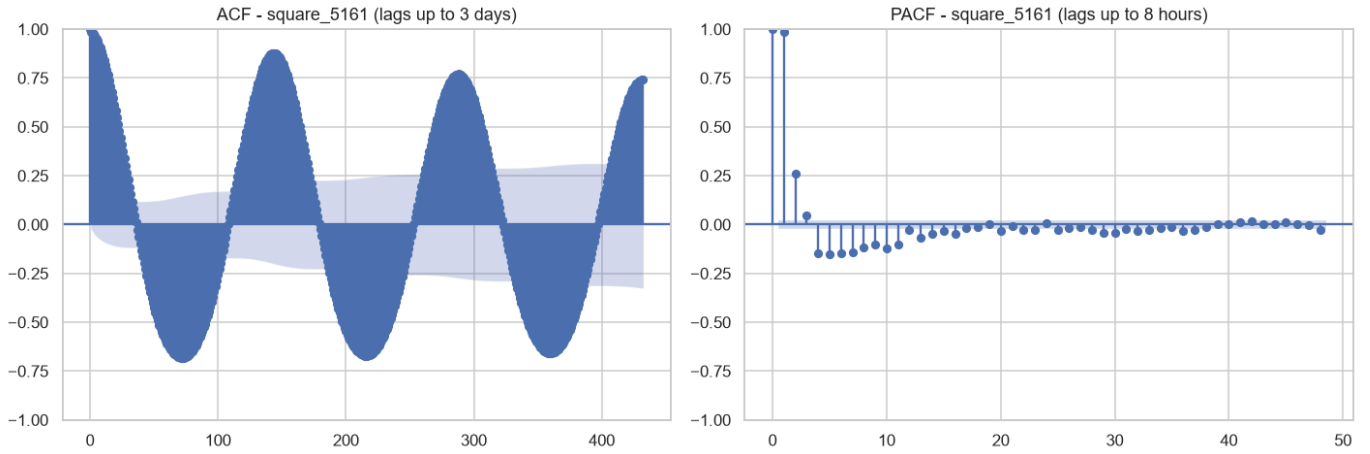


Figure 5. Seasonal component and ACF diagnostics used to motivate explicit temporal structure.

Implication for modelling. The EDA supports models that can use recent observations while representing recurring daily/weekly structure. This is why the statistical model uses Fourier terms, while the neural models receive a one-day sequence plus cyclical time features. The spatial differences also justify evaluating all three busiest squares rather than relying on one area.

Methodology: Forecasting Setup, Models and Tuning

Experimental design. One model instance was trained for each model type and each of the three busiest squares, producing nine model–square combinations. The common training period is Nov 1–Dec 15, while Dec 16–22 is the held-out test week (1,008 steps). Forecasts are strictly one step ahead and walk forward through the test period using the true newly observed value after each prediction. The models never feed their own predictions back as the next input.

Input representation and preprocessing. LSTM and TCN inputs contain a one-day history (144 ten-minute steps). Each time step contains five features: standardized traffic, sine/cosine time-of-day, and sine/cosine day-of-week. The traffic scaler is fitted using training data only. The chronological split prevents test leakage. Neural hyperparameters were selected on Square 5161 using November for training and Dec 1–15 for validation; the test week remained untouched during tuning.

Model 1 — SARIMA-style model. Because EDA showed strong daily and weekly repetition, the implementation represents those seasonal components using Fourier terms and fits a non-seasonal ARIMA model to the remaining structure. A direct seasonal SARIMAX search with $s = 144$ was initially attempted but was stopped after 3,600 seconds because the state-space computation was too expensive. The practical model therefore used dynamic harmonic regression with ARIMA errors and walk-forward updating.

Model 2 — LSTM. The recurrent network uses one LSTM layer with 64 hidden units, learning rate 0.005, Adam optimisation, MSE loss, batch size 64 and 20 epochs. The selected 144-step window corresponds to one day, matching the dominant autocorrelation period. The selected

model has 18,241 parameters.

Model 3 — TCN. The TCN uses six residual blocks with dilations 1, 2, 4, 8, 16 and 32, causal weight-normalized convolutions, ReLU, dropout and residual skip paths. The selected configuration uses 16 channels per block, dropout 0.2, kernel size 3 and learning rate 0.001. Its 253-step receptive field is longer than the 144-step input window, and it contains 9,185 trainable parameters.

Tuning strategy. SARIMA harmonic counts and ARIMA orders were selected by AIC on the training period for each square. LSTM and TCN settings were selected by validation MSE on Square 5161 and then reused for the other squares. This is deliberately more conservative than tuning against the test week and avoids giving the neural models square-specific test information.

Methodology: Hyperparameter Evidence and Results

SARIMA tuning. Stage 1 compared daily harmonics 2, 4, 6 and 8 with weekly harmonics 2, 4 and 6. More harmonics generally reduced AIC. The selected settings were 6 daily/6 weekly for 5059 and 8 daily/6 weekly for 5161 and 5259. Stage 2 compared four ARIMA orders; (2,1,1) had the lowest AIC for every square.

Square	Daily H	Weekly H	Best ARIMA	AIC
5059	6	6	(2,1,1)	85844.4
5161	8	6	(2,1,1)	87964.3
5259	8	6	(2,1,1)	82619.2

LSTM tuning. Hidden size and learning rate were varied first. The best validation MSE was 0.0227 with 64 hidden units and learning rate 0.005. Adding a second layer, reducing the window to 72, or extending it to 288 all worsened validation MSE. The 288-step option also took about ten times longer to train in the tuning run (351 s versus 36 s).

Setting	Hidden	LR	Layers	Window	Val MSE
32/0.001	32	0.001	1	144	0.0291
32/0.005	32	0.005	1	144	0.0259

64/0.001	64	0.001	1	144	0.0243
Selected	64	0.005	1	144	0.0227
128/0.001	128	0.001	1	144	0.0235
128/0.005	128	0.005	1	144	0.0250
2 layers	64	0.005	2	144	0.0269
72-step	64	0.005	1	72	0.0357
288-step	64	0.005	1	288	0.0272

TCN tuning. The best validation MSE was also 0.0227, with 16 channels and dropout 0.2. Kernel size 5 and a learning rate of 0.003 were worse. One kernel-5/learning-rate-0.003 run took 1,061 s, while the other candidates took 83–256 s; background system load was not investigated further.

Setting	Channels	Dropout	Kernel	LR	Val MSE
16/0.1	16	0.1	3	0.001	0.0242
Selected	16	0.2	3	0.001	0.0227
32/0.1	32	0.1	3	0.001	0.0249
32/0.2	32	0.2	3	0.001	0.0228
Kernel 5	16	0.2	5	0.001	0.0437
LR .003	16	0.2	3	0.003	0.0265
Kernel5 + LR	16	0.2	5	0.003	0.0381

Timing and hardware. All timing values use Python `time.perf_counter()` and represent a single run, not an average. Hardware was Intel64 Family 6 Model 186 Stepping 2, approximately 7.7 GB RAM, CPU only and no GPU.

Model	Phase	5161	5059	5259
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SARIMA	Grid search (15 fits)	47.1	65.7	63.6
SARIMA	Inference	54.9	52.6	52.0
LSTM	Training (20 epochs)	60.8	61.7	311.5
LSTM	Inference	0.06	0.07	0.09
TCN	Training (20 epochs)	130.4	113.0	121.3
TCN	Inference	0.10	0.09	0.09

Results and Discussion

Forecast accuracy. The final test-week results are shown below. SARIMA has the lowest MAE, MAPE and RMSE on all three busiest squares under this experimental setup. The neural models remain relatively close, particularly on 5059 and 5259.

Square	Model	MAE	MAPE %	RMSE
5161	SARIMA	82.70	9.43	122.78
5161	LSTM	98.87	17.15	129.17
5161	TCN	87.92	9.98	130.90
5059	SARIMA	69.54	7.67	97.82
5059	LSTM	73.58	7.85	101.36
5059	TCN	78.99	9.04	106.07
5259	SARIMA	68.65	9.10	92.86
5259	LSTM	71.82	9.77	96.86
5259	TCN	72.31	10.42	96.11

Forecast plots. The nine original actual-versus-predicted plots are retained below because together they show the consistency of the daily cycle, the overnight behaviour, the late-week decline, and the individual model deviations that the summary metrics cannot show.

Timing interpretation. SARIMA grid searches took 47.1–65.7 s per square and test-week inference took about 52–55 s. LSTM training took 60.8, 61.7 and 311.5 s, while inference took 0.06–0.09 s. TCN training took 113.0–130.4 s and inference about 0.09–0.10 s. These values show that neural models are very fast at inference once trained, whereas SARIMA's sequential walk-forward updating is substantially slower at test time.

Results, Conclusion, Limitations and Future Work

Cross-area interpretation. Square 5161 is the most challenging case for the neural models: LSTM records MAE 98.87 and MAPE 17.15%, while TCN records MAE 87.92 and MAPE 9.98%. On 5059 and 5259, the neural models are closer to SARIMA. The seasonal-strength values are 0.858 for 5161, 0.891 for 5059 and 0.497 for 5259. These values do not by themselves explain the model differences. One plausible explanation is that the SARIMA model explicitly represents weekday/weekend variation through weekly Fourier terms, but this was not tested as a causal explanation.

Error analysis. The high LSTM MAPE on 5161 is partly visible in overnight troughs, where predictions remain around 250–300 while actual traffic is closer to 150–200. MAPE divides by the actual value, so relatively small absolute errors become large percentages when traffic is low. The TCN on 5161 has the highest RMSE (130.90); its worst day by daily MAE is Dec 17, when it under-predicts a midday peak by roughly 500 traffic units. The analysis identifies the miss but does not establish why that peak occurred.

Effect of tuning. With an earlier, smaller Fourier grid, SARIMA MAE was 81.9, 70.6 and 71.7 for 5161, 5059 and 5259. After tuning, the values became 82.7, 69.5 and 68.7. Thus lower AIC did not improve every test result. The TCN change from 32 to 16 channels improved MAE from 97.8 to 87.9 on 5161, 87.6 to 79.0 on 5059, and 72.6 to 72.3 on 5259. The LSTM configuration did not change and its final results remained the same.

Conclusion. This study compared SARIMA, LSTM and TCN for one-step-ahead Internet traffic forecasting on the three busiest squares of the Milan CDR grid. Under the specified train/test split and walk-forward procedure, SARIMA produced the lowest MAE, MAPE and RMSE in all three areas. The result is dataset- and experiment-specific: it does not establish that SARIMA will always outperform neural models. The neural approaches were trained on roughly 6,300

windows per square, tuning was performed on one square, and the test period was one week.

Limitations. Each model was trained once per square with one random seed. Timing measurements were single runs and varied noticeably, including an LSTM training time of about 312s on one square versus about 61s on the other two. Neural hyperparameters were selected only on 5161, the SARIMA harmonic grid ended at eight daily and six weekly harmonics, and each square was modelled independently rather than jointly.

Future work. I will be repeating the experiments with several random seeds and multiple test weeks—including holidays—would test stability. The SARIMA harmonic grid could be extended and day-of-week indicators could be compared directly with Fourier terms. A spatio-temporal model could use neighbouring squares as additional inputs, motivated by the observed correlations but designed to learn which neighbours are actually informative. Other extensions include attention/Transformer models, CNN-LSTM or TCN variants, richer exogenous variables, probabilistic forecasting and longer evaluation periods.

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